Justice by Design: Auditing Machine Learning in Suspect Profiling.

Transcript.

Hello everyone, my name is Matilda Nilsson, and today I'm excited to present my research proposal titled "Justice by Design: Auditing Machine Learning in Suspect Profiling." This proposal builds upon my earlier literature review, which explored how traditional offender profiling is evolving under the influence of artificial intelligence and machine learning. In this talk, I'll guide you through my research idea, starting with the problem space, the questions and objectives, and the key literature grounding this work. Then I'll outline the methodology, ethical considerations, and proposed artefact that this project will produce.
Finally, I'll conclude with the timeline and anticipated contributions to both scholarship and practice.

Let's begin with a bit of context. For decades, suspect profiling has been a cornerstone of investigative practice used to infer offender characteristics from crime scenes, behavioral evidence, and victim interactions.
This approach, often rooted in psychology, has produced some famous case studies but also sparked criticism for its subjectivity and lack of reproducibility.
In other words, much of it relied on *intuition* and intuition can be biased. Now, in the last decade, machine learning has entered the scene with promises of objectivity. Algorithms can process massive amounts of historical data from criminal records and surveillance logs to social media patterns spotting trends far beyond human capacity. It sounds ideal: faster, data-driven, consistent.

But this raises a central question: If algorithms are trained on biased data, can they ever truly be neutral? And if not, how do we ensure that fairness and accountability are not lost in the pursuit of efficiency? These questions form the ethical and analytical heart of this research.

The core problem is one of trust and transparency. Machine learning promises a level of precision and consistency that human profilers could never achieve. But it also risks reproducing historical bias at a scale never before possible.
A well-known example is the COMPAS risk-assessment algorithm used in the U.S. justice system. Angwin et al. (2016) found that COMPAS disproportionately labelled Black defendants as “high risk,” even when their reoffending rates were lower than those of white defendants. This illustrates how *data reflects the society it comes from*. If that society is unequal, so too are its algorithms.

Another issue is opacity or the “black box” problem. Many machine-learning systems are proprietary, complex, and poorly documented. Even experts can struggle to explain how an algorithm reached its conclusion.
In a field as sensitive as criminal justice, this is not just a technical concern, it’s a moral one. So, the gap I’ve identified is this: while there are numerous frameworks for algorithmic fairness in general AI, very few directly address the specific evidentiary, ethical, and social stakes of suspect profiling.
This project seeks to close that gap by developing a tailored audit framework to evaluate fairness and accountability in machine-learning-based profiling.

This brings us to the guiding research question: How can machine-learning approaches to suspect profiling be critically evaluated to ensure fairness, accountability, and ethical compliance within criminal-justice applications?

The aim of this study is not to build a new profiling algorithm, but to critically assess existing ones. By examining how profiling models are trained, validated, and deployed, I aim to identify best practices and potential red flags for ethical compliance. Ultimately, this project aims to contribute to what I call *"justice by design"* the deliberate integration of ethical safeguards into AI systems before harm can occur.

To achieve this aim, the study pursues five objectives:

1. Critically review existing literature on traditional and ML-based profiling.
2. Identify key ethical dimensions, particularly fairness, accountability, and transparency: the FAT/AI principles.
3. Design a methodology for auditing machine-learning profiling systems.
4. Illustrate the framework through an applied example or conceptual case.
5. Evaluate implications for future research, governance, and practice.

Together, these objectives create a coherent path from *theory* → *analysis* → *application*. This structure also reflects the philosophy of responsible computing, not just asking whether we *can* automate profiling, but whether we *should*, and under what conditions. The foundation of this research rests on three distinct but interlinked strands of literature.

First, the psychological tradition. *Kocsis (2004)* and *Kapardis (2024)* describe how early profiling was influenced by behavioral science and investigative psychology. However, these methods were often intuitive, unverified, and dependent on individual expertise, leading to inconsistent results across investigators.

Second, the technological turn. Studies such as *Bokolo, Chen and Liu (2023)* and *Dakalbab et al. (2022)* showcase how machine learning and deep learning can classify offenders or predict criminal behavior using structured datasets. These models enhance efficiency, but they raise new issues: they often operate as black boxes, and their training data may replicate systemic bias.

Third, the ethical and legal critique. *Almeida, McStay and Sutcliffe (2021)* argue that facial-recognition technologies blur the line between surveillance and privacy violation. Meanwhile, the *European Parliament (2020)* highlights the challenges the GDPR poses to AI, particularly around lawful data processing and algorithmic accountability. When synthesized, these three strands expose a paradox: the same data-driven precision that promises progress can also amplify injustice if left unchecked.

From this literature emerge several key debates that shape the field today. First, predictive utility versus fairness. High accuracy does not automatically mean ethical validity. A model may predict behavior well but still discriminate against particular groups. Second, transparency versus opacity. Many machine-learning models, especially neural networks, are so complex that even their creators cannot fully explain their decision logic. This creates challenges for explainability and legal accountability. Finally, automation versus human oversight. Should algorithms inform human decisions, or should they be trusted to act autonomously? And if an AI system makes a discriminatory judgment, who bears responsibility, the developer, the agency, or the data itself?

These debates highlight why my proposed methodology emphasizes interpretation and critical evaluation rather than algorithmic development.
In short, my project asks not *how well* these systems perform, but *whether* their performance can ever be ethically justified.

Let's move now to methodology. This research uses a qualitative interpretivist approach, since the aim is not to produce numerical predictions but to *interpret meaning, bias, and structure*. Phase 1 will conduct a scoping review of academic, technical, and policy literature to extract key criteria for evaluation. Phase 2 will apply a framework-analysis method, comparing existing ML profiling systems, like COMPAS, against these criteria. Phase 3 synthesizes this analysis into a conceptual audit framework. The framework will use both qualitative themes (e.g. ethical rationales, design assumptions) and quantitative fairness indicators like disparate-impact ratios and false-positive rates. This combination allows a balanced perspective, critical yet concrete. By mixing ethical reasoning with computational awareness, this project bridges two worlds that rarely meet: data science and moral philosophy.

Now, every strong research design must also reflect on its own boundaries. Because this study doesn't have access to live law-enforcement data or commercial algorithms, it relies on publicly available, anonymized sources. This means I cannot fully test or replicate certain proprietary systems, but this is a deliberate ethical choice. The focus here is *not* on building or validating a model; it's about asking *how and why* such models make the decisions they do. By staying interpretive, the project avoids reinforcing harmful biases. In future work, this framework could be extended into empirical testing perhaps with anonymized datasets or synthetic data but for now, critical safety takes precedence. It's better to study injustice safely than risk reproducing it accidentally. This reflexive awareness doesn't weaken the project it actually strengthens it by demonstrating ethical maturity and research transparency.

The proposed framework follows a clear, structured process:

1. Catalogue existing ML profiling models and describe their decision contexts.

2. Evaluate bias and fairness metrics, examining precision parity, false-positive parity, or demographic parity.

3. Map those findings against established standards like the GDPR, the BCS Code of Conduct, and FAT/AI principles.

This process creates a three-layer analytical framework:

- *Technical layer*: model mechanics and data structures.

- *Ethical layer*: fairness and accountability checks.
- *Legal layer*: checks compliance with regulation and rights.

Together, they provide a holistic lens through which profiling technologies can be judged.

The key outcome of this project is a conceptual artefact, the Machine-Learning Profiling Audit Framework.

It has four main components:

- Purpose: to evaluate ML profiling systems against ethical and legal benchmarks.
- The Inputs: public datasets, model documentation, and previous audits.
- Process: bias detection, ethical mapping, and structured evaluation.
- Outputs: a visual report or dashboard summarizing fairness scores and risk indicators.

This framework will be designed for researchers, ethics boards, and policy makers. It turns abstract ethical theory into something concrete and usable, a practical bridge between philosophical ideals and computational realities. In short, it's about transforming ethics from a checkbox into a design principle.

Potential risks include reinforcing stereotypes through case selection, misinterpretation of findings, or handling semi-anonymous data.
And to mitigate this study uses only publicly available, anonymized data.

Findings are presented as critical evaluations, never endorsements. Every stage aligns with the GDPR, the Data Protection Act (2018), and the BCS Code of Conduct. Importantly, these safeguards don't just ensure compliance, they enhance credibility. By embedding ethics into the design, the research models the very behavior it advocates. The project will seek institutional ethical approval, classified as *low physical but moderate ethical sensitivity* due to its subject matter.

The proposed timeline spans twelve weeks and follows a clear progression: Each phase will include reflection logs, supervisor feedback, and iterative revision ensuring academic rigor and consistency.

This project contributes in three significant ways. First, it creates a cross-disciplinary bridge integrating insights from criminology, data science, and ethics. Very few studies examine suspect profiling through all three lenses simultaneously. Second, it offers a practical framework for evaluating fairness and transparency in machine-learning systems. Which can guide developers, researchers, and regulatory bodies alike. Third, it advances the conversation on ethical AI governance, showing that accountability is not a limitation but a foundation. By demonstrating that fairness is a feature, not an afterthought, this research promotes a model of justice that's proactive, and not reactive.

Machine learning increasingly shapes who is deemed risky, suspicious, or safe. And yet, if we cannot explain why a system makes that judgment, *can you really call it justice?* This project takes a step toward resolving that tension, by designing a framework that centers fairness, accountability, and transparency as core design principles. It reframes profiling from a predictive task into a moral design challenge, where ethics are engineered into the process rather than added afterthought. In doing so, it contributes to a vision of justice that is both data-driven and deeply human.

Thank you for listening. I welcome any questions or reflections.

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